

Test Data Setup

About Data

Organism	<i>Saccharomyces cerevisiae</i> (Baker's Yeast)
Dataset	Schurch et al. 2016, (PMID 27022035)
Samples	3 wild-type replicates + 3 Snf2 knockout replicates
Layout	Single-end, 50 bp reads; unstranded
Reference	Ensembl Fungi (R64-1-1 / S288C)

Learning Goal

By the end of this handout, trainees should be able to download a small public RNA-seq dataset and prepare the yeast reference genome and annotation.

1. Dataset and Source Citation

The test data used in this training guide come from the (Schurch et al., 2016) RNA-seq benchmarking study. The study was designed to answer a practical experimental design question: how many biological replicates are needed for reliable RNA-seq differential expression analysis, and which analysis tools perform best under different replicate numbers.

The full study generated a large yeast RNA-seq experiment with biological replicates from two conditions. For training, this pipeline uses a small subset of six samples so that the complete workflow can run quickly on a laptop while preserving a realistic experimental structure.

Citation

Schurch, N. J., Schofield, P., Gierliński, M., Cole, C., Sherstnev, A., Singh, V., Wrobel, N., Gharbi, K., Simpson, G. G., Owen-Hughes, T., Blaxter, M., & Barton, G. J. (2016). How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use? *RNA*, 22(6), 839–851. <https://doi.org/10.1261/rna.053959.115>

Data Accession

Condition	Sample ID
Wild Type (WT)	ERR458493, ERR458494, ERR458495
Snf2 Knockout (KO)	ERR458496, ERR458497, ERR458498
ENA Study	ERP004763

Prerequisites

Before downloading the data, it is important to first create a local environment for the analysis.

For this, we provide a file in the scripts folder named environment.yaml. This file creates a local Conda environment called rnaseq-pipeline, which contains all the required dependencies, including the software and packages needed to run the RNA-seq analysis.

```
cd path/to/environment      #This command changes the path to dependency file  
conda env create -f environment.yaml  
conda activate rnaseq-pipeline
```

2. Downloading dataset

For this step we will be doing the following steps

- Make a directory(folder) to store the raw data, reference genome and index files
- Install the required tools
- Download RNA-seq sample (only 1 file for training purposes; follow script for all files)
- Compress the FASTQ files
- Download the reference genome and decompress
- Download annotation file and decompress
- Check downloaded files

```
mkdir -p data/raw references # Creates separate folder for reference genome files  
conda install -c bioconda sra-tools pigz -y # Installs packages to download raw seq files  
# Command to download raw sequence file  
fasterq-dump --outdir ./ \  
--threads 4 \  
--skip-technical \  
--progress ERR458493
```

```

pigz -f ./ERR458493.fastq # Compress the raw sequence file to save storage

cd data/references # Change directory to reference directory

# Command to download reference genome file

curl -L
https://ftp.ensemblgenomes.ebi.ac.uk/pub/current/fungi/fasta/saccharomyces\_cerevisiae/dna/Saccharomyces\_cerevisiae.R64-1-1.dna.toplevel.fa.gz -o
./genome.fa.gz

gzip -d ./genome.fa.gz # Decompress reference genome

# Command to download annotation file

curl -L -f
https://ftp.ensemblgenomes.ebi.ac.uk/pub/current/fungi/gtf/saccharomyces\_cerevisiae/Saccharomyces\_cerevisiae.R64-1-1.62.gtf.gz -o annotation.gtf.gz

gzip -d ./annotation.gtf.gz # Decompress annotation file

ls -lh data/raw references # Check downloaded files

```

File type	Role
FASTQ files	Raw sequencing reads
genome.fa	Reference genome sequence
annotation.gtf	Gene location and structure information